

GATE CSE: 5. Operating Systems - 6. Miscellaneous

1. [GATE 1995 — 1 Mark]

A linker is given object modules for a set of programs that were compiled separately. What information need to be included in an object module?

- A. Object code
- B. Relocation bits
- C. Names and locations of all external symbols defined in the object module
- D. Absolute addresses of internal symbols

2. [GATE 1997 — 1 Mark]

When an interrupt occurs, an operating system

- A. ignores the interrupt
- B. always changes state of interrupted process after processing the interrupt
- C. always resumes execution of interrupted process after processing the interrupt
- D. may change state of interrupted process to 'blocked' and schedule another process

3. [GATE 2005 — 1 Mark]

A user level process in Unix traps the signal sent on a Ctrl-C input, and has a signal handling routine that saves appropriate files before terminating the process. When a Ctrl-C input is given to this process, what is the mode in which the signal handling routine executes?

- A. kernel mode
- B. super user mode
- C. privileged mode
- D. user mode

4. [GATE 2005 — 1 Mark]

A student wishes to create symbolic links in a computer system running Unix. Three text files named “file1”, “file2” and “file3” exist in her current working directory, and the student has read and write permissions for all three files. Assume that file1 contains information about her hobbies, file2 contains information about her friends and file3 contains information about her courses. The student executes the following sequence of commands from her current working directory. Which of the following types of information would be lost from her file system? I. Hobbies II. Friends III. Courses

```
ln -s file1 file2
ln -s file2 file3
```

- A. I and II only
- B. II and III only
- C. II only
- D. I and III only

5. [GATE 2005 — 1 Mark]

The shell command `find . -name passwd -print` is executed in `/etc` directory of a computer system running Unix. Which of the following shell commands will give the same information as the above command when executed in the same directory?

- A. `ls passwd`
- B. `cat passwd`
- C. `grep name passwd`
- D. `grep print passwd`

6. [GATE 2007 — 2 Mark]

Consider n jobs $J_1, J_2, \dots, J_n$ such that job J_i has execution time t_i and a non-negative integer weight w_i . The weighted mean completion time of the jobs is defined to be $\frac{\sum_{i=1}^n w_i T_i}{\sum_{i=1}^n w_i}$ where T_i is the completion time of job J_i . Assuming that there is only one processor available, in what order must the jobs be executed in order to minimize the weighted mean completion time of the jobs?

- A. Non-decreasing order of t_i
- B. Non-increasing order of w_i
- C. Non-increasing order of $w_i t_i$
- D. Non-increasing order of $\frac{w_i}{t_i}$